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United States Patent Application Publication

20250279682

Kind Code

A1

Publication Date

September 04, 2025

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ELECTRIC-MOTOR STATOR COOLING STRUCTURE AND ELECTRIC MOTOR

Abstract

Disclosed are an electric-motor stator cooling structure and an electric motor. The electric-motor stator cooling structure includes a first sealing oil cover and a second sealing oil cover. The first sealing oil cover is provided with an oil intake hole; the second sealing oil cover is provided with an oil output hole; the first sealing oil cover and the second sealing oil cover are respectively used for covering two opposite ends of a stator winding, such that the first sealing oil cover and one end face of a stator core enclose a first cavity structure, which is in communication with the oil intake hole, and the second sealing oil cover and the other end face of the stator core enclose a second cavity structure, which is in communication with the oil output hole; and a cooling-oil channel is provided inside the stator core.

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Appl. No.: 19/213038

Filed: May 20, 2025

Foreign Application Priority Data

CN

202310336992.4

Mar. 31, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2024/078941 20240228 PENDING child US 19213038

Publication Classification

Int. Cl.: H02K1/20 (20060101); H02K3/487 (20060101); H02K5/10 (20060101)

U.S. Cl.:

CPC H02K1/20 (20130101); H02K3/487 (20130101); H02K5/10 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2024/078941, filed on Feb. 28, 2024, which claims priority to Chinese Patent Application No. 202310336992.4, filed on Mar. 31, 2023. All of the aforementioned applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present application relates to the technical field of electric motors, and more particularly, to an electric-motor stator cooling structure and an electric motor.

BACKGROUND

[0003] With the development of new energy vehicles, the electric motor gradually develops in the direction of small volume and high power density, which puts forward higher requirements on the heat dissipation capacity of the electric motor. There are multiple existing cooling methods for electric motors, among which the most effective is oil-immersion cooling of the stator windings. Specifically, oil covers are used to encapsulate both ends of the stator windings, forming sealed chambers that allow the ends of the stator windings to be fully immersed in cooling oil for thermal dissipation. However, the above-described cooling method can only perform heat exchange on the ends of stator windings, and there may be a case where the heat exchange is not uniform throughout the stator assembly, and the cooling effect on the stator assembly as a whole is limited.

SUMMARY

[0004] The problem to be solved by the present application is how to enable the whole stator assembly to perform effective heat exchange and uniformly cool the stator assembly.

[0005] In order to solve the above-mentioned problems, the present application provides an electric-motor stator cooling structure including a first sealing oil cover and a second sealing oil cover, wherein the first sealing oil cover is provided with an oil intake hole; the second sealing oil cover is provided with an oil output hole; the first sealing oil cover and the second sealing oil cover are respectively used for covering two opposite ends of a stator winding, such that the first sealing oil cover and one end face of a stator core enclose a first cavity structure, which is in communication with the oil intake hole, and the second sealing oil cover and the other end face of the stator core encloses a second cavity structure, which is in communication with the oil output hole; and a cooling-oil channel is provided inside the stator core, and is in communication with the first cavity structure and the second cavity structure.

[0006] The technical effect of the present application is that the first sealing oil cover and the second sealing oil cover may be respectively covered at opposite ends of the stator winding, such that the first sealing oil cover and one end face of the stator core enclose a first cavity structure, and one end of the stator winding is located in the first cavity structure, and the second sealing oil cover and the other end face of the stator core enclose a second cavity structure, and the other end of the stator winding is located in the second cavity structure, and the cooling-oil channel inside the stator core is in communication with the first cavity structure and the second cavity structure, so that the first cavity structure, the cooling-oil channel, and the second cavity structure may form a complete

sealed cavity. When the electric motor needs to be cooled, the cooling oil can flow into the first cavity structure from the oil intake hole, then the cooling oil flows into the stator core through the cooling-oil channel, and the cooling oil can flow into the second cavity structure along the cooling-oil channel so that the cooling oil can flow out from the oil output hole after passing through the first end of the stator winding, the stator core and the tail end of the stator winding successively, thereby enabling the whole stator assembly to contact the cooling oil, i.e. the whole stator assembly can perform effective heat exchange cooling, and the stator assembly can be uniformly cooled.

[0007] Preferably, the first sealing oil cover includes a first oil cover body, a first inner ring portion and a first outer ring portion, the first inner ring portion and the first outer ring portion are respectively connected to the first oil cover body, and the first inner ring portion, the first outer ring portion and the first oil cover body form a first annular groove structure, an open end of the first annular groove structure is used to abut against an end face of the stator core so as to enclose the first cavity structure, and the second sealing oil cover includes a second oil cover body, a second inner ring portion and a second outer ring portion, wherein the second inner ring portion and the second outer ring portion are respectively connected to the second oil cover body, and the second inner ring portion, the second outer ring portion and the second oil cover body form a second annular groove structure, and an open end of the second annular groove structure is used to abut against an end face of the stator core to enclose the second cavity structure.

[0008] Preferably, the electric-motor stator cooling structure further includes a sealing ring mounted on an end face of the stator core, an end of the sealing ring facing the stator core being provided with a plurality of protruding teeth spaced apart, the plurality of protruding teeth being adapted to be inserted into a plurality of slot openings of the stator core in a one-to-one correspondence, and the sealing ring being provided between the stator core and the first inner ring portion.

[0009] Preferably, the electric-motor stator cooling structure further includes a sealing slot wedge for being provided inside the slot opening, an end of the sealing slot wedge is provided with a male stop structure, and a portion of the protruding teeth inserted into the slot opening is provided with a female stop structure, and the male stop structure is for being connected to the female stop structure.

[0010] Preferably, one end of the sealing ring facing away from the stator core is provided with a first stop structure, the first inner ring portion is provided with a second stop structure matching the first stop structure, and the second stop structure is connected to the first stop structure.

[0011] Preferably, the electric-motor stator cooling structure further includes a first sealing gasket provided at a junction of the first stop structure and the second stop structure.

[0012] Preferably, the electric-motor stator cooling structure further includes a second sealing gasket, wherein the first outer ring portion is provided with a mounting groove, and an open end of the mounting groove is adapted to abut against an end face of the stator core, and the second sealing gasket is provided in the mounting groove.

[0013] Preferably, the first outer ring portion is provided with an oil inlet groove on a side thereof facing away from the first inner ring portion, and the first outer ring portion is provided with the oil intake hole, and the oil intake hole is in communication with the oil inlet groove.

[0014] The present application also provides an electric motor including a motor housing and an electric-motor stator cooling structure as described above, a first sealing oil cover of the electric-motor stator cooling structure is provided in the electric motor housing, the electric motor housing is provided with an oil inlet, the first sealing oil cover and a side wall of the electric motor housing enclose an oil inlet cavity structure, and the oil inlet is in communication with the oil inlet cavity structure.

[0015] The advantageous effects of the electric motor according to the present application are the same as those of the electric-motor stator cooling structure, which will not be described in detail herein.

[0016] Preferably, the second sealing oil cover of the electric-motor stator cooling structure is provided in the electric motor housing, the electric motor housing is provided with an oil outlet cavity structure, the oil output hole of the second sealing oil cover is in communication with the oil outlet cavity structure, and the electric motor housing is provided with an oil outlet, the oil outlet is in communication with the oil outlet cavity structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a schematic view showing the internal structure of an electric-motor stator cooling structure according to an embodiment of the present application.

[0018] FIG. 2 is a partially enlarged schematic view of FIG. 1.

[0019] FIG. 3 is a schematic view showing the structure of the first sealing oil cover according to an embodiment of the present application.

[0020] FIG. 4 is a schematic view showing the structure of the second sealing oil cover according to an embodiment of the present application.

[0021] FIG. 5 is a schematic view showing the structure of the sealing ring according to an embodiment of the present application.

[0022] FIG. 6 is a schematic view showing the structure of the sealing slot wedge according to an embodiment of the present application.

[0023] FIG. 7 is a partially enlarged schematic view of the stator core according to an embodiment of the present application.

[0024] FIG. 8 is a schematic view showing the structure of a stator winding and a stator core according to an embodiment of the present application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0025] In order that the above-described objects, features, and advantages of the present application are attained and can be understood in detail, a more particular description of embodiments of the application, briefly summarized above, may be had by reference to the appended drawings.

[0026] It is noted that the terms “first”, “second”, and the like in the description and the claims of the present application and in the above-described drawings, are used for distinguishing between similar objects and not necessarily for describing a particular sequential or chronological order. It is to be understood that the data so used are interchangeable under appropriate circumstances such that the embodiments of the application described herein are capable of operation in sequences other than those illustrated or otherwise described herein.

[0027] With reference to FIGS. 1, 3, 4 and 8, an electric-motor stator cooling structure according to an embodiment of the present application includes a first sealing oil cover 1 and a second sealing oil cover 2, wherein the first sealing oil cover 1 is provided with an oil intake hole 3; the second sealing oil cover 2 is provided with an oil output hole 4; the first sealing oil cover 1 and the second sealing oil cover 2 are respectively used for covering two opposite ends of a stator winding 5, such that the first sealing oil cover 1 and one end face of a stator core 6 enclose a first cavity structure 7, the first cavity structure 7 is in communication with the oil intake hole 3, and the second sealing oil cover 2 and the other end face of the stator core 6 enclose a second cavity structure 8, the second cavity structure 8 is in communication with the oil output hole 4; and a cooling-oil channel 9 is provided inside the stator core 6, and the cooling-oil channel 9 is in communication with the first cavity structure 7 and the second cavity structure 8.

[0028] Specifically, the electric-motor stator cooling structure may include a first sealing oil cover 1, which may be provided with an oil intake hole 3, and a second sealing oil cover 2, which may be provided with an oil output hole 4. In the present embodiment, the first sealing oil cover 1 may be covered at one end of the stator winding 5, that is, one end of the stator winding 5 is located inside

the first sealing oil cover **1**, so that the first sealing oil cover **1** and one end face of the stator core **6** enclose the first cavity structure **7**, and one end of the stator winding **5** is located inside the first cavity structure **7**, and the oil intake hole **3** may be in communication with the first cavity structure **7**.

[0029] Similarly, the second sealing oil cover **2** may be covered on the other end of the stator winding **5**, that is, the other end of the stator winding **5** is located inside the second sealing oil cover **2**, so that the second sealing oil cover **2** and the other end face of the stator core **6** may enclose the second cavity structure **8**, and the other end of the stator winding **5** is located inside the second cavity structure **8**, and the oil output hole **4** may be in communication with the second cavity structure **8**. The inside of the stator core **6** may be provided with a cooling-oil channel **9**, and in this embodiment, the stator core **6** is provided with a plurality of cooling-oil channels **9**, and the plurality of cooling-oil channels **9** are arranged at intervals along the circumferential direction of the stator core **6**, the cooling-oil channels **9** are arranged to extend in the axial direction of the stator core **6**, and the cooling-oil channels **9** may be in communication with the first cavity structure **7** and the second cavity structure **8**.

[0030] When the electric motor needs to be cooled, the cooling oil can flow into the first cavity structure **7** from the oil intake hole **3**, then the cooling oil flows into the stator core **6** through the cooling-oil channel **9**, and the cooling oil can flow into the second cavity structure **8** along the cooling-oil channel **9** so that the cooling oil can flow out from the oil output hole **4** after passing through the head end of the stator winding **5**, the stator core **6** and the tail end of the stator winding **5** successively, thereby enabling the whole stator assembly to contact the cooling oil, i.e. the whole stator assembly can perform effective heat exchange cooling, and the stator assembly can be uniformly cooled.

[0031] Referring to FIGS. **1**, **3** and **4**, in some embodiments, the first sealing oil cover **1** includes a first oil cover body **101**, a first inner ring portion **102** and a first outer ring portion **103**, the first inner ring portion **102** and the first outer ring portion **103** are respectively connected to the first oil cover body **101**, and the first inner ring portion **102**, the first outer ring portion **103** and the first oil cover body **101** form a first annular groove structure **104**, an open end of the first annular groove structure **104** is used to abut against an end face of the stator core **6** so as to enclose the first cavity structure **7**, and the second sealing oil cover **2** includes a second oil cover body **201**, a second inner ring portion **202** and a second outer ring portion **203**, wherein the second inner ring portion **202** and the second outer ring portion **203** are respectively connected to the second oil cover body **201**, and the second inner ring portion **202**, the second outer ring portion **203** and the second oil cover body **201** form a second annular groove structure, and an open end of the second annular groove structure is used to abut against an end face of the stator core **6** to enclose the second cavity structure **8**.

[0032] Specifically, the first sealing oil cover **1** may include a first oil cover body **101**, a first inner ring portion **102**, and a first outer ring portion **103**, wherein the first oil cover body **101** is shaped as an annular structure, the first inner ring portion **102** may be connected to an inner ring edge of the first oil cover body **101**, the first outer ring portion **103** may be connected to an outer ring edge of the first oil cover body **101**, and the first inner ring portion **102** and the first outer ring portion **103** are both located on the same side of the first oil cover body **101** so that the first oil cover body **101**. The first inner ring portion **102** and the first outer ring portion **103** may form a first annular groove structure **104**, the first outer ring portion **103** is provided with an oil intake hole **3**, and a plurality of oil intake holes **3** may be provided at intervals along the circumferential direction of the first outer ring portion **103**.

[0033] The first annular groove structure **104** can match the annular structure formed by the end coils of the stator winding **5** so that the first annular groove structure **104** can cover an end of the stator winding **5** and wrap the end coils of the stator winding **5**; the open end of the first annular groove structure **104** can abut against the end face of the stator core **6**; in the present embodiment,

the end of the first outer ring portion **103** away from the first oil cover body **101** bears against the end face of the stator core **6**; and the end of the first inner ring portion **102** away from the first oil cover body **101** is connected to the stator core **6** via a sealing member to form a sealing structure. Thus enclosing the first cavity structure **7**, cooling oil can flow within the first annular groove structure **104** to cool an end of the stator winding **5**.

[0034] In some embodiments, one end of the first outer ring portion **103** remotes from the first oil cover body **101** may be provided with a snap-fit structure, wherein the snap-fit structure includes a plurality of snap-fits, and the plurality of snap-fits are arranged at intervals along the circumferential direction of the first outer ring portion **103**; the stator core **6** may be provided with a snap-fit groove structure; and when the first outer ring portion **103** abuts against the stator core **6**, the snap-fit structure may be inserted into the snap-fit groove structure, so that the first outer ring portion **103** and the stator core **6** are connected more firmly.

[0035] Similarly, the second sealing oil cover **2** may include a second oil cover body **201**, a second inner ring portion **202**, and a second outer ring portion **203**, wherein the shape of the second oil cover body **201** is an annular structure, the second inner ring portion **202** may be connected to an inner ring edge of the second oil cover body **201**, the second outer ring portion **203** may be connected to an outer ring edge of the second oil cover body **201**, and the second inner ring portion **202** and the second outer ring portion **203** are both located on the same side of the second oil cover body **201** so that the second oil cover body **201**. The second inner ring portion **202** and the second outer ring portion **203** may form a second annular groove structure, the second oil cover body **201** may be provided with an oil output hole **4**, and a plurality of oil output holes **4** may be provided at intervals along the circumferential direction of the second oil cover body **201**.

[0036] The second annular groove structure may be covered on the other end of the stator winding **5**, and the coil at the end of the stator winding **5** is wrapped around; the open end of the second annular groove structure may abut against the end face of the stator core **6**; in the present embodiment, one end of the second outer ring portion **203** away from the second oil cover body **201** abuts against the end face of the stator core **6**, and one end of the second inner ring portion **202** away from the second oil cover body **201** is connected to the stator core **6** via a sealing member to form a sealing structure, thereby enclosing the second cavity structure **8**; and the cooling oil may flow in the second annular groove structure to cool the other end of the stator winding **5**.

[0037] Referring to FIGS. **2**, **5**, and **7**, in some embodiments, the electric-motor stator cooling structure further includes a sealing ring **10** mounted on an end face of the stator core **6**, an end of the sealing ring **10** facing the stator core **6** being provided with a plurality of protruding teeth **11** spaced apart, the plurality of protruding teeth **11** being adapted to be inserted into a plurality of slot openings **12** of the stator core **6** in a one-to-one correspondence, and the sealing ring **10** being provided between the stator core **6** and the first inner ring portion **102**.

[0038] Specifically, the electric-motor stator cooling structure may further include a sealing ring **10**, wherein the sealing ring **10** may be mounted on an end face of the stator core **6**, an end of the sealing ring **10** facing the stator core **6** may be provided with a plurality of protruding teeth **11** at intervals; in the present embodiment, the end face of the stator core **6** may be provided with a plurality of slot openings **12**, the plurality of slot openings **12** are evenly spaced along the circumference of the stator core **6**, and the plurality of protruding teeth **11** may be inserted into the plurality of slot openings **12** of the stator core **6** in a one-to-one correspondence manner so that the sealing ring **10** is firmly mounted on the end face of the stator core **6**. The sealing ring **10** may be provided between the stator core **6** and the first inner ring portion **102**, and the sealing ring **10** may form a sealing structure between the stator core **6** and the first inner ring portion **102**, so that the sealing ring **10** may increase the sealability of the first cavity structure **7**, and may prevent the cooling oil from leaking from between the first inner ring portion **102** and the stator core **6**.

[0039] In the other embodiments, a sealing ring **10** may be provided between the stator core **6** and the second inner ring portion **202**, and the sealing ring **10** may increase the sealability of the second

cavity structure **8**.

[0040] Referring to FIGS. **2**, **5**, and **6**, in some embodiments, the electric-motor stator cooling structure further includes a sealing slot wedge **13** for being provided inside the slot opening **12**, and an end of the sealing slot wedge **13** is provided with a male stop structure **131**, and a portion of the protruding teeth **11** inserted into the slot opening **12** is provided with a female stop structure **111**, and the male stop structure **131** is for being connected to the female stop structure **111**.

[0041] Specifically, the electric-motor stator cooling structure may further include a sealing slot wedge **13**, and in the present embodiment, a plurality of sealing slot wedges **13** may be provided, and the plurality of sealing slot wedges **13** may be provided inside the plurality of slot openings **12** in a one-to-one correspondence manner, and an end of the sealing slot wedge **13** may be provided with a male stop structure **131**, and a part of the protruding teeth **11** inserted into the slot openings **12** may be provided with a female stop structure **111**, and the male stop structure **131** may be matingly connected to the female stop structure **111**. In other embodiments, the sealing slot wedge **13** may be provided with a female stop structure **111** at an end thereof, and a male stop structure **131** may be provided at a portion where the protruding teeth **11** are inserted into the slot opening **12**.

[0042] The sealing slot wedge **13** itself has the function of increasing the sealability to prevent the cooling oil from leaking out of the slot opening **12**, and the sealing slot wedge **13** and the protruding teeth **11** are connected by the mating of the male stop structure **131** and the female stop structure **111** to further increase the sealability to prevent the cooling oil from leaking out of the slot opening **12**.

[0043] Preferably, the insulating paint can be dropped into the slot opening **12**, and the space gap of the sealing slot wedge **13**, the protruding teeth **11** of the sealing ring **10**, and the stator core **6** can be filled with the insulating paint, so that the inside of the stator core **6** forms a sealed whole, and the cooling oil can be effectively prevented from leaking.

[0044] Referring to FIGS. **2** and **3**, in some embodiments, the end of the sealing ring **10** facing away from the stator core **6** is provided with a first stop structure **14**, and the first inner ring portion **102** is provided with a second stop structure **15** matching the first stop structure **14**, and the second stop structure **15** is connected to the first stop structure **14**.

[0045] Specifically, the end of the sealing ring **10** facing away from the stator core **6** may be provided with a first stop structure **14**, the first stop structure **14** being provided with one turn around the circumference of the sealing ring **10**, and the end of the first inner ring portion **102** facing the stator core **6** may be provided with a second stop formation **15** matching the first stop structure **14**, the second stop formation **15** also being provided with one turn around the circumference of the first inner ring portion **102**.

[0046] In the present embodiment, an end of the first inner ring portion **102** close to the stator core **6** is recessed in a direction facing away from the first outer ring portion **103** to form a second stop structure **15**, the first stop structure **14** can cooperate with the second stop structure **15**, and an end of the first stop structure **14** can abut against the first inner ring portion **102**, so that the second stop structure **15** is connected to the first stop structure **14**, and the sealability of the first cavity structure **7** can be further increased by providing the first stop structure **14** and the second stop structure **15**.

[0047] Referring to FIG. **2**, in some embodiments, the electric-motor stator cooling structure further includes a first sealing gasket **16** disposed at the junction of the first and second stop structures **14**, **15**.

[0048] Specifically, the electric-motor stator cooling structure may further include a first sealing gasket **16**, wherein the first sealing gasket **16** may be provided at the junction of the first stop structure **14** and the second stop structure **15**; in the present embodiment, the second stop structure **15** is formed by the recess of one end of the first inner ring portion **102** in a direction facing away from the first outer ring portion **103**, and the distance between the second stop structure **15** and the first outer ring portion **103** is greater than the radial dimension of the stator core **6**; and when the

first sealing oil cover **1** is covered on the stator winding **5**, the end face **102** of the first inner ring portion is in a suspended state, namely, the end face of the second stop structure **15** is in a suspended state.

[0049] In the present embodiment, the first sealing gasket **16** may be interposed between the first stop structure **14** and the second stop structure **15**, and more clearly, the first sealing gasket **16** is provided between the side wall of the first stop structure **14** and the side wall of the second stop structure **15**, so that the first sealing gasket **16** can increase the sealability of the junction between the first stop structure **14** and the second stop structure **15**, and can further prevent the leakage of the cooling oil.

[0050] Preferably, a groove structure may be provided on a side wall of the first stop structure **14**, and a protrusion structure may be provided on a side of the first sealing gasket **16** facing the first stop structure **14**, and the protrusion structure may be inserted into the groove structure so that the first sealing gasket **16** may be more firmly provided at the junction of the first stop structure **14** and the second stop structure **15**.

[0051] Referring to FIGS. **1** and **2**, in some embodiments, the electric-motor stator cooling structure further includes a second sealing gasket **17**, wherein the first outer ring portion **103** is provided with a mounting groove **1031**, and an open end of the mounting groove **1031** is used to abut against an end face of the stator core **6**, and the second sealing gasket **17** is provided in the mounting groove **1031**.

[0052] Specifically, the electric-motor stator cooling structure may further include a second sealing gasket **17**, an end of the first outer ring portion **103** facing the stator core **6** may be provided with a mounting groove **1031**, and an open end of the mounting groove **1031** may abut against an end face of the stator core **6**, and the second sealing gasket **17** may be provided in the mounting groove **1031**, so that the second sealing gasket **17** may increase sealability between the first outer ring portion **103** and the end face of the stator core **6**, and may prevent cooling oil from leaking between the first outer ring portion **103** and the stator core **6**.

[0053] Referring to FIGS. **1** to **3**, in some embodiments, an oil inlet groove **19** is provided on a side of the first outer ring portion **103** facing away from the first inner ring portion **102**, and the first outer ring portion **103** is provided with the oil intake hole **3**, which is in communication with the oil inlet groove **19**.

[0054] Specifically, a side of the first outer ring portion **103** facing away from the first inner ring portion **102** may be provided with an oil inlet groove **19** which is formed by the first outer ring portion **103** being recessed in a direction close to the first inner ring portion **102**, and the oil inlet groove **19** may be provided to extend in a circumferential direction of the first outer ring portion **103**, the first outer ring portion **103** may be provided with an oil intake hole **3**, and the oil intake hole **3** may be in communication with the oil inlet groove **19**, and more clearly, a groove bottom of the oil inlet groove **19** may be provided with the oil intake hole **3**, and cooling oil may first enter the oil inlet groove **19** and then flow into the first cavity structure **7** from the oil intake hole **3**, so that the cooling oil may uniformly flow into the first cavity structure **7**.

[0055] Referring to FIGS. **1** to **3**, there is also provided a motor according to another embodiment of the present application, which includes a motor housing **20** in which a first sealing oil cover **1** of the electric-motor stator cooling structure is provided, the electric motor housing **20** being provided with an oil inlet **21**, and a motor stator cooling structure as described above, the first sealing oil cover **1** and a side wall of the electric motor housing **20** defining an oil inlet cavity structure **22**, the oil inlet **21** is in communication with the oil inlet cavity structure **22**.

[0056] Specifically, the electric motor may include a motor housing **20** and a motor stator cooling structure, the first sealing oil cover **1** of the electric-motor stator cooling structure may be provided inside the electric motor housing **20**, and the electric motor housing **20** may be provided with an oil inlet **21**, the first sealing oil cover **1** may be provided with an oil inlet groove **19**, and an open end of the oil inlet groove **19** may abut against a side wall of the electric motor housing **20**, so that the

oil inlet groove **19** may enclose an oil inlet cavity structure **22** with the side wall of the electric motor housing **20**, and the oil inlet **21** may be in communication with the oil inlet cavity structure **22**.

[0057] When the electric motor needs to be cooled, the cooling oil can be pumped to the oil inlet **21** on the electric motor housing **20** via the oil pump, the cooling oil can flow into the oil inlet cavity structure **22** from the oil inlet **21**, the cooling oil then flows into the first cavity structure **7** via the oil intake hole **3** to cool one end of the stator winding **5**, then passes through the cooling-oil channel **9** inside the stator core **6** to perform heat exchange and dissipation on the stator core **6**, and finally flows into the second cavity structure **8** to cool the other end of the stator winding **5**, and the cooling oil flows out from the oil output hole **4**, so that the whole stator assembly can be efficiently and thermally dissipated.

[0058] Referring to FIG. **1**, in some embodiments, the second sealing oil cover **2** of the electric-motor stator cooling structure is provided in the electric motor housing **20**, the electric motor housing **20** is provided with an oil outlet cavity structure **23**, the oil output hole **4** of the second sealing oil cover **2** is in communication with the oil outlet cavity structure **23**, and the electric motor housing **20** is provided with an oil outlet port **24**, the oil outlet port **24** is in communication with the oil outlet cavity structure **23**.

[0059] Specifically, the second sealing oil cover **2** of the electric-motor stator cooling structure can be provided inside the electric motor housing **20**, the electric motor housing **20** can also be provided with an oil outlet cavity structure **23**, the oil output hole **4** on the second sealing oil cover **2** can be in communication with the oil outlet cavity structure **23**, the electric motor housing **20** can also be provided with an oil outlet **24**, the oil outlet **24** can be in communication with the oil outlet cavity structure **23**, the cooling oil can be collected in the oil outlet cavity structure **23** after flowing out from the second cavity structure **8**, and the cooling oil can uniformly flow out from the oil outlet **24** so that the cooling oil can be conveniently collected, and after processing the cooling oil, entering the next cooling cycle, the cooling efficiency can be increased.

[0060] In some embodiments, a side surface of the second outer ring portion **203** of the second sealing oil cover **2** may be provided with a second sealing groove, an open end of the second sealing groove being arranged in the direction facing the electric motor housing **20**, and the open end of the second sealing groove may abut against a side wall of the electric motor housing **20**, and the electric-motor stator cooling structure may also be provided with a third sealing gasket, and the third sealing gasket may be arranged in the second sealing groove, so as to increase the sealability between the second outer ring portion **203** and the electric motor housing **20**.

[0061] Similarly, a first seal groove may be provided on a side surface of the first outer ring portion **103** of the first sealing oil cover **1**, and an open end of the first seal groove may be provided in the direction facing the electric motor housing **20**, and the open end of the first seal groove may abut against a side wall of the electric motor housing **20**, and a third sealing gasket may be provided in the first seal groove, so that the sealability between the first outer ring portion **103** and the electric motor housing **20** may be increased.

[0062] Although the present disclosure has been described above, the scope of the present disclosure is not limited thereto. It will be apparent to those skilled in the art that various modifications and variations can be made in the present application without departing from the spirit or scope of the disclosure.

Claims

1. An electric-motor stator cooling structure, comprising a first sealing oil cover and a second sealing oil cover, wherein the first sealing oil cover is provided with an oil intake hole; the second sealing oil cover is provided with an oil output hole; the first sealing oil cover and the second sealing oil cover are respectively used for covering two opposite ends of a stator winding, such that

the first sealing oil cover and one end face of a stator core enclose a first cavity structure, the first cavity structure is in communication with the oil intake hole, and the second sealing oil cover and the other end face of the stator core enclose a second cavity structure, the second cavity structure is in communication with the oil output hole; and a cooling-oil channel is provided inside the stator core, and the cooling-oil channel is in communication with the first cavity structure and the second cavity structure.

2. The electric-motor stator cooling structure according to claim 1, wherein the first sealing oil cover comprises a first oil cover body, a first inner ring portion and a first outer ring portion, the first inner ring portion and the first outer ring portion are respectively connected to the first oil cover body, and the first inner ring portion, the first outer ring portion and the first oil cover body form a first annular groove structure, an open end of the first annular groove structure is used to abut against an end face of the stator core so as to enclose the first cavity structure; and the second sealing oil cover comprises a second oil cover body, a second inner ring portion and a second outer ring portion, the second inner ring portion and the second outer ring portion are respectively connected to the second oil cover body, and the second inner ring portion, the second outer ring portion and the second oil cover body form a second annular groove structure, and an open end of the second annular groove structure is used to abut against an end face of the stator core to enclose the second cavity structure.

3. The electric-motor stator cooling structure according to claim 2, further comprising a sealing ring mounted on an end face of the stator core, an end of the sealing ring facing the stator core being provided with a plurality of protruding teeth spaced apart, the plurality of protruding teeth being adapted to be inserted into a plurality of slot openings of the stator core in a one-to-one correspondence, and the sealing ring being provided between the stator core and the first inner ring portion.

4. The electric-motor stator cooling structure according to claim 3, further comprising a sealing slot wedge for being provided inside the slot opening, wherein an end of the sealing slot wedge is provided with a male stop structure, and a portion of the protruding teeth inserted into the slot opening is provided with a female stop structure, and the male stop structure is for being connected to the female stop structure.

5. The electric-motor stator cooling structure according to claim 3, wherein one end of the sealing ring facing away from the stator core is provided with a first stop structure, and the first inner ring portion is provided with a second stop structure matching the first stop structure, and the second stop structure is connected to the first stop structure.

6. The electric-motor stator cooling structure according to claim 5, further comprising a first sealing gasket provided at a junction of the first stop structure and the second stop structure.

7. The electric-motor stator cooling structure according to claim 2, further comprising a second sealing gasket, wherein the first outer ring portion is provided with a mounting groove, and an open end of the mounting groove is adapted to abut against an end face of the stator core, and the second sealing gasket is provided in the mounting groove.

8. The electric-motor stator cooling structure according to claim 2, wherein the first outer ring portion is provided with an oil inlet groove on a side thereof facing away from the first inner ring portion, and the first outer ring portion is provided with the oil intake hole, and the oil intake hole is in communication with the oil inlet groove.

9. An electric motor, comprising a motor housing and the electric-motor stator cooling structure according to claim 1, wherein a first sealing oil cover of the electric-motor stator cooling structure is provided in the electric motor housing, the electric motor housing is provided with an oil inlet, the first sealing oil cover and a side wall of the electric motor housing enclose an oil inlet cavity structure, and the oil inlet is in communication with the oil inlet cavity structure.

10. The electric motor according to claim 9, wherein the second sealing oil cover of the electric-motor stator cooling structure is provided in the electric motor housing, the electric motor housing

is provided with an oil outlet cavity structure, the oil output hole of the second sealing oil cover is in communication with the oil outlet cavity structure, and the electric motor housing is provided with an oil outlet, the oil outlet is in communication with the oil outlet cavity structure.
